# AYUSHI RAJ

**Software Developer — Full-Stack & AI**

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## PROFESSIONAL SUMMARY

Full-Stack Developer building web applications, browser extensions, and AI tools with React, Next.js, TypeScript, and Python. Co-built **AstreWork** (astrework.app), shipping its live resume-scoring studio and the **AstrePilot** Chrome extension (Manifest V3) for one-click job application autofill. Cares about clean UI, fast page loads, and practical security.

## TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C#, SQL, HTML5/CSS3
- **Frontend & UI:** React 19, Next.js 16 (App Router), Redux, Zustand, Tailwind CSS, Monaco Editor
- **Backend & APIs:** Node.js, Express.js, FastAPI, .NET Core Web API, REST APIs
- **AI & Browser Automation:** Google GenAI SDK, Multi-LLM Routing, Chrome Extensions (MV3), Shadow DOM, Playwright, Puppeteer
- **Databases & Caching:** Turso LibSQL, SQLite, PostgreSQL, MongoDB, Redis 7, Prisma ORM, ChromaDB
- **Security, Testing & Cloud:** Web Crypto API (AES-GCM, PBKDF2), HMAC-SHA256, JWT, Vitest, PyTest, xUnit, Docker, Git

## TECHNICAL PROJECTS

**AstreWork — AI Career Platform & AstrePilot Extension (Next.js 16, React 19, Chrome MV3, Gemini API)** **Live Demo | GitHub**

- **Co-Developer & Lead Frontend Architect:** Built AstreWork's client-side ATS match engine and Gemini-powered scoring studio in Next.js 16, grading live job posts in under a second.
- **AstrePilot Chrome Extension (MV3):** Created an isolated Shadow-DOM HUD with HMAC-SHA256 token auth that autofills Greenhouse, Lever, and Ashby forms in ~15 seconds.
- **Workspace & Print Engine:** Built the React 19 and Zustand workspace with live resume-audit checks and a single-page A4 print layout with zero layout shift.

**Astra Vision — Interactive Code Sandbox & AST Visualizer (React, Monaco, FastAPI, ChromaDB)** **Live Demo | GitHub**

- **In-Browser IDE:** Built a browser-based code analysis studio pairing Monaco Editor with an interactive dependency-graph canvas for multi-file repos.
- **Live AST Inspection:** Connected streaming FastAPI and Tree-Sitter endpoints to highlight syntax nodes, call paths, and sandbox output in under 200ms.
- **Semantic Code Search:** Created a natural-language repository search UI backed by ChromaDB vector embeddings and isolated execution previews.

**Alyra Lock — Client-Side Zero-Knowledge Password Vault (React, TypeScript, Web Crypto, Express)** **Live Demo | GitHub**

- **Client-Side Vault UI:** Built a zero-knowledge React password manager that derives master keys strictly in browser memory so raw credentials never hit the server.
- **Memory-Safe Crypto State:** Wired Web Crypto API (AES-GCM 256-bit, 100,000-iteration PBKDF2) into ephemeral React state that wipes decrypted keys on idle lock.
- **Fast Encrypted Sync:** Shipped instant client-side vault search, password strength checks, and JWT-authenticated Express sync completing in under 200ms.

## TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

**Commonwealth Bank (CommBank) — Software Development Virtual Simulation** **June 2026**

- **Problem & Scope:** Diagnosed and resolved silent data-overwrite bugs across high-traffic C#/.NET Core financial API controllers backed by MongoDB.
- **Action Taken:** Extended C# Web API controllers with $set atomic operators for partial updates and modernized the React/Redux Goal Manager UI component.
- **Engineering Outcome:** Authored automated test suites using xUnit and Moq covering boundary conditions to guarantee transactional consistency.

**Y Combinator (YC Startup - Shiptivity) — Software Development Virtual Simulation** **June 2026**

- **Problem & Scope:** Fixed key-constraint collisions and slow re-rendering on a real-time drag-and-drop Kanban workflow board.
- **Action Taken:** Architected SQLite priority reordering logic using atomic transactions and integrated Dragula drag-and-drop state hooks in React.
- **Engineering Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL build bottlenecks in legacy Node.js Webpack configurations.

**Walmart USA — Advanced Software Development Virtual Simulation** **May 2026**

- **Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory queues fed by unnormalized legacy spreadsheets.
- **Action Taken:** Implemented a generic K-ary Max Heap in Java using bitwise shift operators (<<, >>>) and designed 3NF relational schemas.
- **Engineering Outcome:** Accelerated queue indexing by 35% and built automated Python ETL pipelines (csv, sqlite3) to consolidate disparate spreadsheets.

## EDUCATION

**Bachelor of Computer Applications (BCA) — Amity University Noida** **2025 – 2028 (Full-Time Remote Available)**

- **Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)